

QUESTION 2

16 MARKS

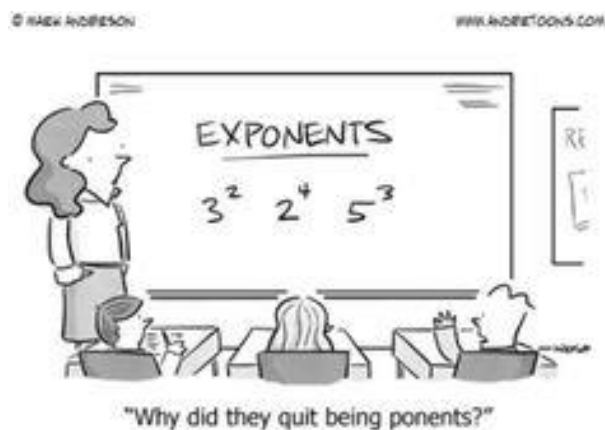
a) Simplify fully, without the use of a calculator, showing all working:

1) $\frac{10^a - 10^{a-1}}{10^{a-2}}$ (4)

$$\begin{aligned}
 &= \frac{10^a - 10^a \cdot 10^{-1}}{10^a \cdot 10^{-2}} \\
 &= \frac{\cancel{10^a} (1 - 10^{-1})}{\cancel{10^a} \cdot 10^{-2}} \quad \checkmark \quad \checkmark \quad \checkmark \quad A \\
 &= 10^2 - 10 \quad \checkmark \quad CA \\
 &= 90 \quad \checkmark \quad CA
 \end{aligned}$$

2) $2\sqrt{8x^{12}} - 4x^2\sqrt{32x^8}$ (3)

$$\begin{aligned}
 &= 4\sqrt{2}x^6 - 16\sqrt{2}x^2 \cdot x^4 \quad \checkmark \quad A \\
 &= 4\sqrt{2}x^6 - 16\sqrt{2}x^6 \quad \checkmark \quad CA \\
 &= -12\sqrt{2}x^6 \quad \checkmark \quad CA
 \end{aligned}$$



QUESTION 7**13 MARKS**

a) The sum of the first n terms of a sequence is given by $S_n = 3^{n+1} - 6$.

1) Determine the sum of the first 6 terms. (2)

$$S_6 = 3^7 - 6 = 2181 \quad \checkmark^m \quad A$$

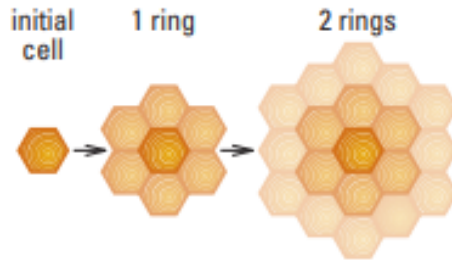
2) Determine the value of the 6th term. (3)

$$S_5 = 3^6 - 6 = 723 \quad \checkmark^A$$

$$T_6 = 2181 - 723 \quad \checkmark^m$$

$$= 1458 \quad \checkmark^{CA}$$

- b) Domestic bees make their honeycomb by starting with a single hexagonal cell, then forming ring after ring of hexagonal cells around the initial cell, as shown.



Source: USDA's Carl Hayden Bee Research Lab)

The numbers of cells in **successive rings** forms an arithmetic sequence, i.e.:

0 ; 6 ; 12 ;etc.

- 1) Determine the number of hexagons in the 15th ring. (3)

$$\begin{aligned}
 T_n &= 6(n-1) \checkmark^A \\
 &= 6(14) \checkmark^m \\
 &= 84 \text{ hexagons} \checkmark^{CA}
 \end{aligned}$$

- 2) What is the total number of cells in the honeycomb after the 50th ring is formed, if the initial cell is excluded? (5)

$$\begin{aligned}
 a &= 0 \checkmark^* \quad n = 50 \checkmark^A \\
 S_{50} &= \frac{50}{2} [2(0) + 6(49)] \\
 &= 7350 \text{ cells} \checkmark^m \checkmark^A \checkmark^{CA}
 \end{aligned}$$